Top quark FCNC anomalous couplings at the future collider FCC-ee

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In this study, we explore the potential of the future circular electron-positron collider (FCC-ee) in searching for Flavor-Changing Neutral Current (FCNC) interactions involving top quarks. Specifically, we investigate the production of single top quarks associated with a light quark ($q = u, c$) in FCNC processes. The analysis relies on Monte Carlo (MC) simulations conducted using the official FCC framework. The simulations are performed for two suggested center-of-mass energies at the FCC-ee: 240 GeV and 365 GeV, corresponding to planned integrated luminosities of 5 ab^{-1} and 1.5 ab^{-1}, respectively. The event selections is made considering both the leptonic decay (electron and muon) and the fully hadronic decay of the top quark in the final state. By employing the CLs method, upper limits at a 95% confidence level (CL) are determined for the sets of branching ratios associated with the decay of the top quark. The statistical combination of results obtained from the leptonic channel, considering both the center-of-mass energy and electron (muon) channel, reveals that at the FCC-ee, upper limits for the branching fractions of $Br(t \to c\gamma) < 1.99 \times 10^{-5}$ and $Br(t \to cZ) < 8.19 \times 10^{-6}$ could be achieved at a 95% CL. In the case of the fully hadronic analysis, the branching ratio after the statistical combination of center-of-mass energies for the processes under investigation are found to be at the order of $Br(t \to c\gamma) < 7.65 \times 10^{-5}$ and $Br(t \to cZ) < 3.28 \times 10^{-5}$.

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I. INTRODUCTION

Flavor Changing Neutral Current (FCNC) interactions of the top quark are not allowed at tree level and are highly suppressed at the loop level due to the Glashow-Iliopoulos-Maiani mechanism (GIM) [3]. The Standard Model (SM) predicts extremely low rates for FCNC decays of the top quark to gluons, photons, Z and Higgs bosons, or up and charm quarks, with expected values at the order of $\mathcal{O}(10^{-10} - 10^{-15})$. These predictions are currently beyond the sensitivity of present and even future experimental capabilities. Hence, any observed deviation from this expected behavior would serve as evidence for the presence of New Physics (NP) beyond the Standard Model (SM) [16]. Furthermore, upcoming high-precision and high-luminosity electron-positron colliders like the FCC-ee are specifically designed to scrutinize the SM with exceptional precision, providing an opportunity to probe and explore physics beyond the SM.

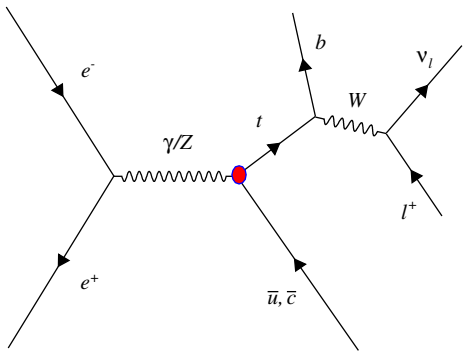

Figure 1: Representative leading-order Feynman diagrams for the production of top quark via FCNC in leptonic decay channel of the W-boson from top decay.

In this study, we aim to assess the sensitivity and potential for the discovery of the FCC-ee in detecting signal events related to Flavor Changing Neutral Current (FCNC) transitions of the top quark. Our investigation is carried out using a model-independent approach known as the Effective Field Theory (EFT) framework. To achieve this, we adopt the strategy outlined in previous works such as [16, 17]. Specifically, we focus on quantifying the expected sensitivity of the FCC-ee to the FCNC couplings of the top quark with the $tq\gamma$ and tqZ interactions.

In our study, we focus on the signal process $e^-e^+ \to Z\gamma \to t\bar{q}$ ($\bar{t}q$), where the Z boson and photon (γ) produced in the electron-positron collision subsequently decay into a top quark (t) and a light quark (q), or an anti-top quark ($\bar{t}$) and a light antiquark ($\bar{q}$). The light quark q can be either an up or a charm quark.

To investigate the sensitivity and discovery prospects of the FCC-ee, we consider both decay modes of the top quark. The first decay mode involves studying the leptonic decay of the top quark, where we examine the decay involving electrons and muons in the final state. The second decay mode focuses on the fully hadronic decay of the top quark, considering the decays that result in jets of hadrons. By exploring both the leptonic and fully hadronic decay modes of the top quark, we aim to comprehensively assess the potential of the FCC-ee in detecting FCNC interactions involving the top quark.

In the leptonic scenario (as shown in Fig. 1), the top quark decays according to the Standard Model (SM) process, where $t \to W^+b \to \ell^+\nu_\ell b$ and $\bar{t} \to W^-\bar{b} \to \ell^-\bar{\nu}_\ell\bar{b}$. As a result, the final state comprises a charged lepton ℓ, missing energy (neutrino), a b-jet, and a light jet, as depicted in Figure 1.

In the fully hadronic scenario (as shown in Fig. 2), the top quark decays as $t \to W^+b \to jjb$

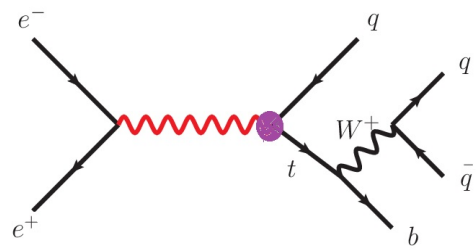

Figure 2: Representative leading order Feynman diagrams for the production of top quark via FCNC in a fully hadronic decay channel.

and $\bar{t} \to W^-\bar{b} \to jj\bar{b}$, where j represents a jet. Consequently, the final state consists of three jets plus an additional light jet, with at least one jet being identified as a b-tagged jet.

Further information regarding the analysis strategy and event selection will be provided in the subsequent sections.

II. THEORETICAL FRAMEWORK

In this section, we outline the theoretical framework and assumptions utilized in our analysis to investigate the FCNC transitions of the top quark at the FCC-ee.

The possibility of the top quark anomalous FCNC couplings with light quarks ($q = u, c$) and a gauge bosons (Z and γ) is explored in a model-independent way considering the most general effective Lagrangian approach [16, 17]. In the search for anomalous FCNC interactions at high-energy colliders, this approach has been extensively studied in the literature for both the lepton and hadron colliders. In this framework, these FCNC vertices are described by higher-dimensional effective operators $\mathcal{L}^{tqX}_{\rm FCNC}$ independently from the underlying theory. Up to dimension-six operators, the FCNC Lagrangian of the tqg, tqH, $tqZ(\sigma^{\mu\nu})$, $tqZ(\gamma_\mu)$ and $tq\gamma$ interactions can be written as [16–18]

$$\begin{aligned}\mathcal{L}^{tqX}_{\rm FCNC} = \sum_{q=u,c} \Bigg[& \frac{g_s}{2m_t}\bar{q}\lambda^a\sigma^{\mu\nu}\left(\zeta^L_{qt}P^L + \zeta^R_{qt}P^R\right)tG^a_{\mu\nu} - \frac{1}{\sqrt{2}}\bar{q}\left(\eta^L_{qt}P^L + \eta^R_{qt}P^R\right)tH \\ & + \frac{g_W}{4c_W m_Z}\bar{q}\sigma^{\mu\nu}\left(\kappa^L_{qt}P_L + \kappa^R_{qt}P_R\right)tZ_{\mu\nu} - \frac{g_W}{2c_W}\bar{q}\gamma^\mu\left(X^L_{qt}P_L + X^R_{qt}P_R\right)tZ_\mu \\ & + \frac{e}{2m_t}\bar{q}\sigma^{\mu\nu}\left(\lambda^L_{qt}P_L + \lambda^R_{qt}P_R\right)tA_{\mu\nu}\Bigg] + \text{h.c.}\,.\end{aligned} \tag{1}$$

In Eq. (1), the dimensionless parameters ζ_{qt}, η_{qt}, κ_{qt}, X_{qt} and λ_{qt} represent the strength of the FCNC interactions of a top quark with gluon, Higgs, Z, and γ, respectively. In the above

equation, q indicates an up or charm quark as well. In this work, we mainly concentrate on the case of top quark FCNC couplings $tq\gamma$ and tqZ at FCCee. At the tree level in the SM, all the above coefficients are zero. In the presence of anomalous FCNC vertices, the straightforward way is to set limits on these coupling's strengths and then translate them to the corresponding branching fractions. In the above equation, g_s is the strong coupling constant, and P_L and P_R denote the left- and right-handed projection operators. In this study, we assume no specific chirality for the anomalous FCNC interactions, and hence, one could set $\kappa_{qt}^L = \kappa_{qt}^R = \kappa_{qt}$, $X_{qt}^L = X_{qt}^R = X_{qt}$ and $\lambda_{qt}^L = \lambda_{qt}^R = \lambda_{qt}$. As we mentioned earlier in the introduction, the single top FCNC signal includes the Z and γ top quark FCNC interactions, and hence, make it possible to study all the coefficients sensitive to the $tq\gamma$ and tqZ.

We should notice here that, the fixed values of $\kappa_{qt} = 0.1$ and $\lambda_{qt} = 0.1$ with $q = u$ and c are chosen for all coupling strengths as the benchmark point if not stated otherwise. Taking these typical benchmarks input for the top FCNC signal, the expected cross sections are calculated.

III. SIMULATED SAMPLES

The analysis is based on the benchmarks introduces in the Future circular collider (FCC) conceptual design in electron-positron collisions at center of mass energy 240GeV and 365 GeV corresponding to the integrated luminosity of $5\,\mathrm{ab}^{-1}$ and $1.5\,\mathrm{ab}^{-1}$, respectively.

The FCC-ee, a proposed high-energy frontier electron-positron collider, holds immense potential for precision measurements in the field of particle physics. In order to capture a wide range of physical processes, events are generated by using various generators. To simulate the response of the detector and reconstruct the events, fast simulation package DELPHES [] is used. DELPHES applies a resolution formula and efficiency to smear the particles generated at the generator level, thus emulating the characteristics of a realistic detector. This complete chain of event generation, simulation, and reconstruction is seamlessly integrated within the Key4HEP software framework. For our analysis, we utilize the official central samples from the Spring2021 campaign, ensuring the reliability and consistency of our results.

The Innovative Detector for an Electron-positron Accelerator (IDEA) is a cutting-edge detector concept designed for future electron-positron colliders like the FCC-ee. It incorporates advanced technologies and design features to achieve exceptional performance in precision measurements, particle identification, and energy resolution. The IDEA detector includes components such as a high-resolution tracking system, electromagnetic and hadronic calorimeters, advanced particle iden-

tification techniques, a magnet system for precise momentum measurement, and forward/backward detectors for comprehensive coverage. The design focuses on optimizing energy and spatial resolution, particle identification efficiency, and overall detector performance across a wide range of particle energies.

A. simulated signal samples

In order to generate the signal events, the FCNC lagrangian introduced in the Eq1, is implemented into the MadGraph5_aMC@NLO event generator by using the `FeynRules` package and the corresponding UFO file. The signal comprises single top production where the FCNC couplings appear at the production level of $e^-e+ \to tq$. In presented cross-sections in the Table I, coupling strengths $\lambda_{tq\gamma}$ and κ_{tqZ} are left free letting only one of them to be non-zero at a time. All samples are generated at leading order (LO).

B. Simulated background samples

In order to estimate the contributions of background processes, `MadGraph5_aMC@NLO`, the `WHIZARD` event generator and parton showering with `PYTHIA6`, and both generation and parton showering with `PYTHIA8` MC generator is used.

MC sample name	Generator	Processes	x-section at 240 GeV [pb]	x-section at 365 GeV [pb]
signal				
mgp8_ee_tbw_FCNC_tua	MadGraph5_aMC@NLO	$e^-e^+ \to tu$	(LO)	
mgp8_ee_tbw_FCNC_tca	MadGraph5_aMC@NLO	$e^-e^+ \to tc$		
mgp8_ee_tbw_FCNC_tuz	MadGraph5_aMC@NLO	$e^-e^+ \to tu$		
mgp8_ee_tbw_FCNC_tcz	MadGraph5_aMC@NLO	$e^-e^+ \to tc$		
background				
p8_ee_ZH	PYTHIA8	$e^-e^+ \to ZH$	202	117
p8_ee_WW	PYTHIA8	$e^-e^+ \to W^+W^-$	16439	10717
p8_ee_ZZ	PYTHIA8	$e^-e^+ \to ZZ$	1360	643
wzp6_egamma_tbnu	WHIZARD+PYTHIA6	$e^+\gamma \to tb\nu$	0.01	0.2
wzp6_egamma_eZ_Zbb	WHIZARD+PYTHIA6	$e^-\gamma \to Zbb$		
wzp6_egamma_eZ_Zqq	WHIZARD+PYTHIA6	$e^-\gamma \to Zqq$		
wzp6_ee_tbjj_singleTop	WHIZARD+PYTHIA6	$e^-e^+ \to Zqq$	-	6
p8_ee_tt	PYTHIA8	$e^-e^+ \to t\bar{t}$	-	800

Table I: MC samples used in this analysis in 240 GeV and 365 GeV center-of-mass-energy

IV. ANALYSIS STRATEGY AND EVENT SELECTION

We conduct our analysis in a model-independent manner, utilizing the effective Lagrangian approach described in Equation (1). Our study focuses on the future collider FCC-ee, operating at center-of-mass energies of $\sqrt{s} = 240$ GeV and 365 GeV. In our analysis, we consider both the leptonic (electron and muon) and fully hadronic decays of the W boson in the top quark decay. To perform our analysis, we utilize centrally produced samples within the FCCSW framework. These samples incorporate an `IDEA` detector response to accurately model the detector effects. We also take into account all major background contributions and thoroughly analyze them. For the discrimination of signals from background processes, we propose a set of kinematic variables as input variables. The specific details of the event selection criteria and kinematic variables will be presented in the following sections. We begin by discussing the leptonic decay channel, followed by a comprehensive presentation of the fully hadronic analysis.

A. Leptonic decay channel

For the analysis of leptonic decay, the signal process is defined as electron-positron annihilation into a Z or photon, which then decays into a top quark and an up or charm quark, along with their antiparticles. The top quark subsequently decays through the Standard Model, producing a W boson and a bottom quark, which further decay into a charged lepton (either an electron or a muon), a corresponding neutrino, and a bottom quark (or its antiparticle). Therefore, the final state consists of a charged lepton, missing energy, a bottom quark, and a light quark. We conduct this analysis separately for muons and electrons. We evaluated various jet clustering algorithms provided in the official framework and ultimately selected the `ee-`k_T `Durham` jet clustering algorithm [13, 14] with the Energy recombination scheme (ES) [19]. Since the final state involves two jets, we specifically consider the exclusive mode with exactly two jets. In the case of the `ee-`k_T `Durham` algorithm, our final results are based on the implementation `JetClustering::clustering_ee_kt(2, 2, 1, 0)` available in the official FCCAnalysis framework. The first argument in `JetClustering::clustering_ee_kt` denotes the exclusive clustering, specifying the number of the event clustered into exactly n-jets. The second argument indicates exactly two jets in the final state. The third argument determines the ordering of jets based on energy, and the last argument corresponds to the E recombination scheme [19].

Based on the expected signature of the signal process, the main background contributing to

the signal 240 GeV center-of-mass energy includes the $e^-e^+ \to WW$, $e^-e^+ \to HZ$, $e^-e^+ \to ZZ$, $e^-e^+ \to tq$, $e^-e^+ \to eebb$ and $e^-e^+ \to eeqq\,(q = u, d, s)$. For the case of 365 GeV center-of-mass energy, in addition to the list mentioned above, the single top production $e^-e^+ \to tbjj$ and top-pair production $e^-e^+ \to t\bar{t}$ will also contribute.

These officially generated signal samples include:

- `p8_ee_ZH_ecm240`
- `p8_ee_WW_ecm240`
- `p8_ee_ZZ_ecm240`
- `wzp6_egamma_tbnu_ecm240`
- `wzp6_gammae_tbnu_ecm240`
- `wzp6_egamma_eZ_Zbb_ecm240`
- `wzp6_gammae_eZ_Zbb_ecm240`
- `wzp6_egamma_eZ_Zqq_ecm240`
- `wzp6_gammae_eZ_Zqq_ecm240`

and the same notation for the 365 GeV, but this time `wzp6_ee_tbjj_singleTop_ecm365` and `p8_ee_tt_ecm365` are also included.

To select events in this study, we impose certain criteria. Specifically, we only consider events that have a single charged lepton (either an electron or a muon) with a total momentum of at least 10 GeV and a polar angle greater than 13 degrees. Additionally, we require the presence of exactly two jets, each with a momentum of at least 20 GeV and a polar angle greater than 13 degrees. Furthermore, one of these jets must be identified as a b-quark jet. To determine the most suitable jet clustering algorithm, we reconstruct the invariant mass distribution of the top quark using different algorithms. The corresponding plot, along with the momentum of the b jet candidate, is displayed in Figure 3.

As previously mentioned, we focus on the `ee-kt Durham` jet clustering algorithm, specifically in conjunction with the `ES` recombination scheme in exclusive mode. Using this combination, we analyze the mass distribution of the reconstructed top quark and the W boson for the $tc\gamma$ signal, as well as the main backgrounds. The corresponding results obtained using the `ee-kt Durham` jet

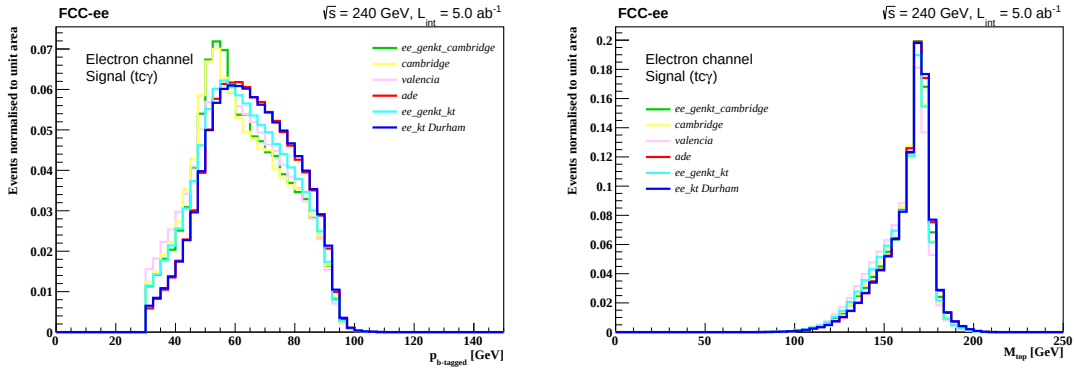

Figure 3: The mass distribution of the reconstructed top quark and the momentum of b jet candidate for $tc\gamma$ signal for different jet clustering algorithms. The distributions are shown for the electron channel and center-of-mass energy of 240 GeV.

clustering algorithm are presented in Figures 4 and 5 for center-of-mass energies of 240 GeV and 365 GeV, respectively.

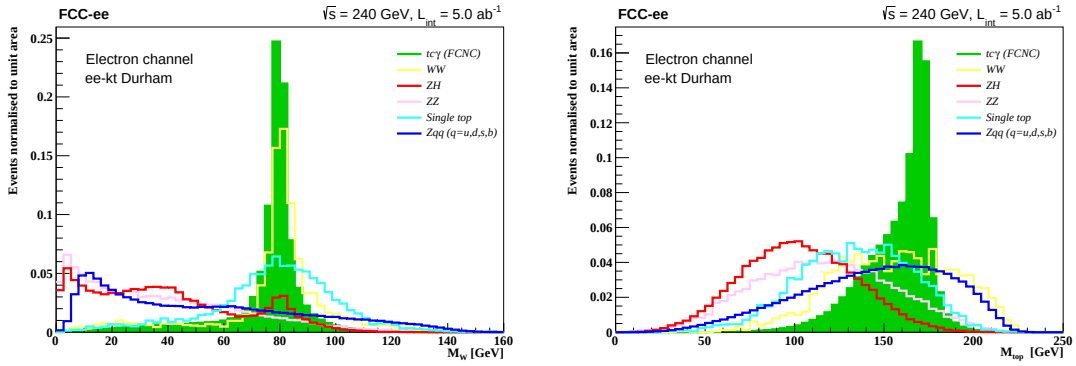

Figure 4: The mass distribution of the reconstructed top quark and the W boson for $tc\gamma$ signal and main backgrounds using the `ee-kt Durham` jet clustering algorithm and for the center-of-mass energy of 240 GeV.

In order to distinguish between signal and background processes effectively, we carefully select variables with the highest discrimination power. The distributions of some of these selected variables are displayed in Figure 6. These distributions correspond to the signal scenario with an anomalous

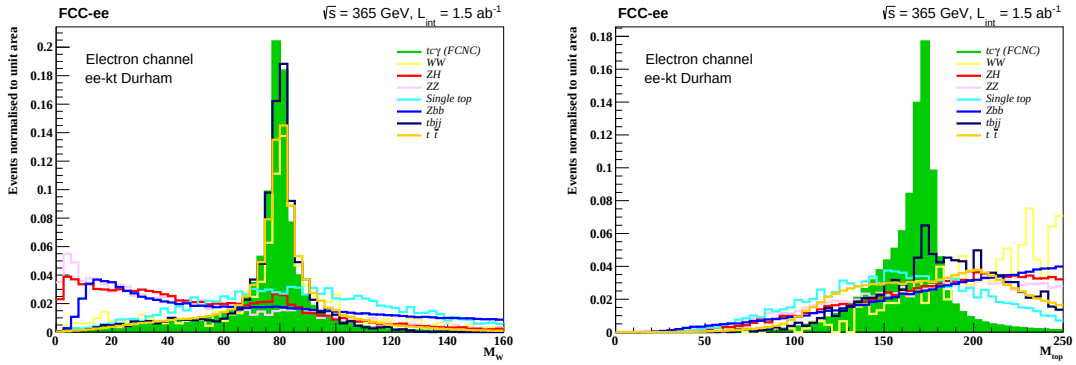

Figure 5: The mass distribution of the reconstructed top quark and the W boson for $tc\gamma$ signal and main backgrounds using the `ee-kt Durham` jet clustering algorithm and for the center-of-mass energy of 365 GeV.

$tc\gamma$ coupling in the electron channel at a center-of-mass energy of 240 GeV. It is worth noting that the same variables are utilized to differentiate the signal from backgrounds in all signal scenarios involving $tc\gamma$ and tcZ. The analyses are conducted separately for the electron and muon channels, as well as for different light quark flavors such as c and u, at center-of-mass energies of both 240 GeV and 365 GeV, while employing the same set of variables.

For the center-of-mass energy of 240 GeV, the following selections are applied.

- Final selection cuts for the case of 240 GeV center-of-mass energy:
  - The reconstructed top-quark mass distribution 140 GeV $< M_{\text{top}} <$ 190 GeV.
  - The reconstructed W boson mass distribution 70 GeV $< M_{\text{W}} <$ 90 GeV.
  - The momentum of b jet candidate $p_{\text{bjet}} >$ 40 GeV.
  - The momentum of the highest energy light jet in each event 40 GeV $< p_{light-jet} <$ 70 GeV.
  - The momentum of electron (or muon) 20 GeV $< p_{\text{electron}} <$ 90 GeV.
  - The Missing Energy 30 GeV $<$ ME $<$ 80 GeV.
  - The angular separation between the lepton and b-jet, $\Delta R(\ell, b - jet) < 3.5$.
  - The cosine of lepton and b-jet, $\cos(\ell, b - jet) < 0.85$

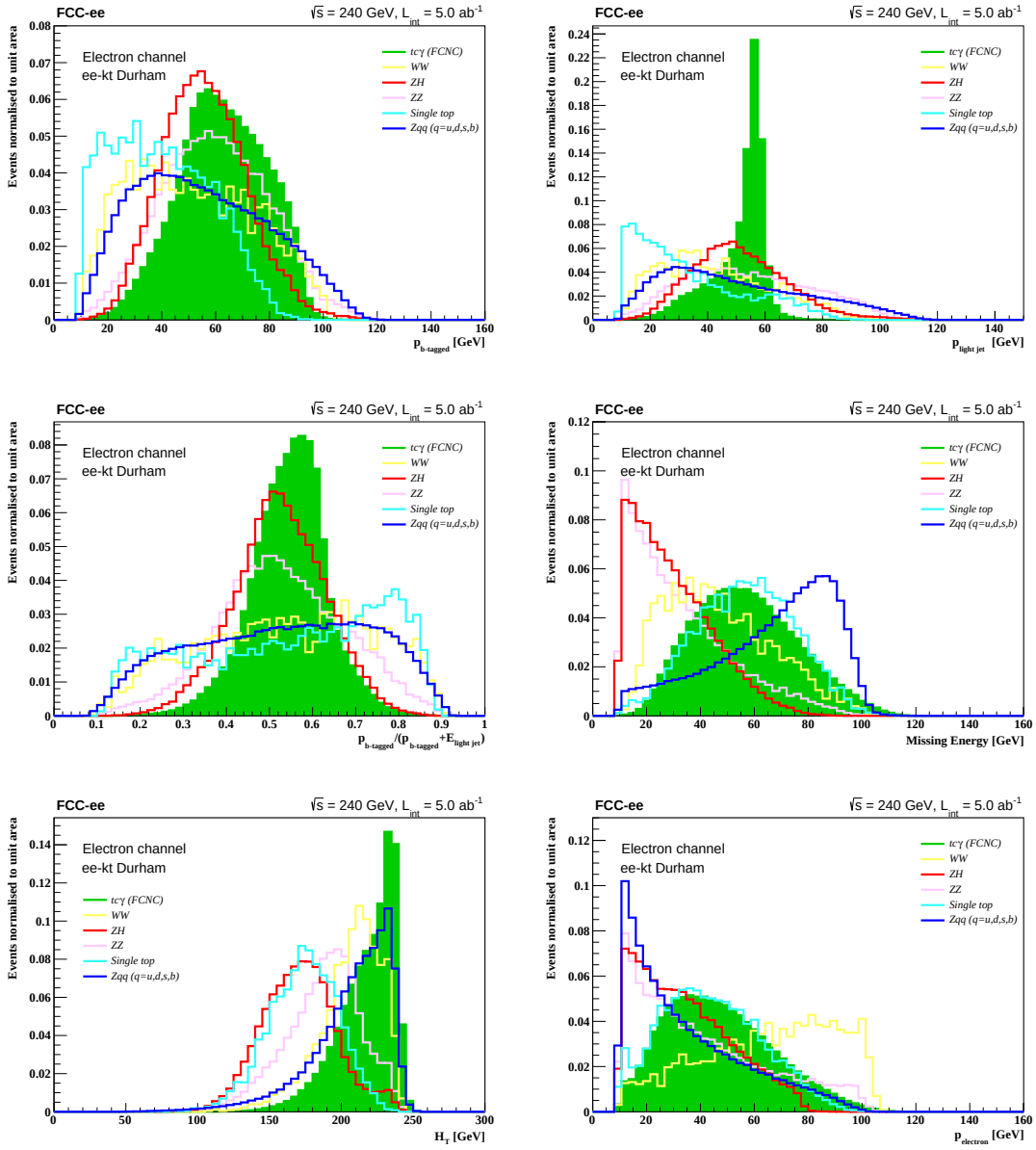

Figure 6: The distributions for the total momentum of a b-tagged jet, the total momentum of light jets, the ratio of the total momentum of b-tagged jets over all jets, the missing energy, the scalar sum of the transverse momentum of all final state objects, and the total momentum of the electron.

- The invariant mass of di-jets, 20 GeV $< M_{(j1,j2)} <$ 150 GeV
- The ratio of total momentum of b-jet to all jets, $0.35 < \frac{p_{\mathrm{b-jet}}}{p_{\mathrm{b-jet}}+p_{\mathrm{light-jet}}} < 0.7$.
- The scalar sum of momentum $H_T > 190$ GeV.

- The total momentum of reconstructed top-quark, 40 GeV $< p^{\rm top} <$ 70 GeV

For the case of 365 GeV, the following selections are considered.

- Final selection cuts for the case of 365 GeV center-of-mass energy:
  - The reconstructed top-quark mass distribution 140 GeV $< M_{\rm top} <$ 190 GeV.
  - The reconstructed W boson mass distribution 70 GeV $< M_{\rm W} <$ 90 GeV.
  - The momentum of b jet candidate 30 GeV $< p_{b-jet} <$ 140 GeV.
  - The momentum of the highest energy light jet in each event 120 GeV $< p_{light-jet} <$ 150 GeV.
  - The momentum of electron (or muon) 20 GeV $< p_{\rm electron} <$ 100 GeV.
  - The Missing Energy 30 GeV $<$ ME $<$ 120 GeV.
  - The angular separation between the lepton and b-jet, $\Delta R(\ell, b-jet) < 3.5$.
  - The cosine of lepton and b-jet, $\cos(\ell, {\rm b-jet}) < 0.85$.
  - The invariant mass of di-jets, 100 GeV $< M_{(j1,j2)} <$ 300 GeV.
  - The ratio of total momentum of b-jet to light jet, $0.2 < \frac{p_{\rm b-jet}}{p_{\rm b-jet}+p_{\rm light-jet}} < 0.5$.
  - The scalar sum of momentum $H_T >$ 300 GeV.
  - The total momentum of reconstructed top-quark, $p^{\rm top} >$ 120 GeV.
  - The cosine of lepton and MET, $\cos(\ell, {\rm MET}) < 0.6$.

Taking the typical pre-selection and final-selection cuts mentioned above, the cross-sections of the SM backgrounds are presented in Table. II and Table. III for 240 and 365 GeV, respectively.

$\sqrt{s}$ = 240 GeV	$tc\gamma$	tcZ	ZH	WW	ZZ	eebb	eeqq (q=uds)	single top
Cross section	14.327	34.132	201.868	16438.5	1358.99	455.7	1635.9	0.010753
Preselection	3.1542	7.5154	1.1953	3.75948	2.76677	19.251	0.155146	0.0003208
Final selection	1.3686	3.2622	0.00847	0.254797	0.01127	0.1807	0.001439	3.2259×10^{-7}

Table II: Cross-sections in the unit of fb for the signal scenarios (electron channel) and for the main SM backgrounds considered in this study passing pre-selection and final selection cuts at a center-of-mass energy of 240 GeV.

$\sqrt{s}$ = 365 GeV	$tc\gamma$	tcZ	ZH	WW	ZZ	$t\bar{t}$	eebb	single top	$tbjj$
Cross section	25.958	50.9517	117.3	10716.5	642.8	800.0	615.2	0.227	5.98
Preselection	5.2107	10.2299	0.6240	2.55696	1.21039	39.7574	25.0175	0.006297	0.2159
Final selection	1.9599	4.3406	0.00410	0.00857	0.00809	1.10752	0.004332	2.276×10^{-6}	0.006947

Table III: Cross-sections in the unit of fb for the signal scenarios (electron channel) and for the main SM backgrounds considered in this study passing pre-selection and final selection cuts at a center-of-mass energy of 365 GeV.

B. Results for the leptonic analysis

The 95% CL upper limits on the top FCNC decays at obtained using the CLs method at the $\sqrt{s}$ = 240 GeV are presented in Table. IV. These limits correspond to the FCCee working at the integrated luminosity of 5 ab^{-1}. The corresponding numbers for the $\sqrt{s}$ = 365 GeV and for an integrated luminosity of 1.5 ab^{-1} are presented in Table. V as well.

$\sqrt{s}$ = 240 GeV	$Br(t \to c\gamma)$	$Br(t \to cZ)$
Electron channel	9.60×10^{-5}	3.53×10^{-5}
Muon channel	6.29×10^{-5}	2.31×10^{-5}

Table IV: The upper limits on the top FCNC decays at 95% C.L obtained using the CLs method at the $\sqrt{s}$ = 240 GeV and for an integrated luminosity of 5 ab^{-1}.

$\sqrt{s}$ = 365 GeV	$Br(t \to c\gamma)$	$Br(t \to cZ)$
Electron channel	1.65×10^{-4}	6.96×10^{-5}
Muon channel	1.47×10^{-4}	6.19×10^{-5}

Table V: Same as Table IV, but this time for the $\sqrt{s}$ = 365 GeV and for an integrated luminosity of 1.5 ab^{-1}.

C. Statistical combination of results in leptonic channel

In Table. VI, the upper limits on the top FCNC decays at 95% C.L extracted using the `Higgs Combine Tool` [20] after the statistical combination of $\sqrt{s}$ = 240 and 365 GeV are presented. After

the statistical combination of energy and electron (muon) channels, the branching ratio of $Br(t \to c\gamma) < 1.99 \times 10^{-5}$ and $Br(t \to cZ) < 8.19 \times 10^{-6}$ are calculated.

$\sqrt{s} = 240$ and 365 GeV	$Br(t \to c\gamma)$	$Br(t \to cZ)$
Energy & channel statistical combination	1.99×10^{-5}	8.19×10^{-6}

Table VI: The upper limits on the top FCNC decays at 95% C.L obtained using the `Higgs combination tools` after statistical combination of $\sqrt{s} = 240$ and 365 GeV, and electron (muon) channel.

D. Fully hadronic channel

In this section, we present the analysis of the FCNC $tq\gamma$ and tqZ couplings in the fully hadronic decay mode of a single top-quark production process. The analysis is conducted at both 240 GeV and 365 GeV center-of-mass energies. For this analysis, we utilize officially generated signal samples, which include:

- `mgp8_ee_fullhad_FCNC_tuz_ecm240`
- `mgp8_ee_fullhad_FCNC_tua_ecm240`
- `mgp8_ee_fullhad_FCNC_tuz_ecm365`
- `mgp8_ee_fullhad_FCNC_tua_ecm365`

For the hadronic samples, we adopt the same configurations that were utilized during the spring 2021 campaign. To perform the jet clustering, we employ the `ee-kt Durham` algorithm in conjunction with the `ES` recombination scheme. Since the final state consists of four jets, we specifically focus on the exclusive mode with exactly four jets. In the case of the `ee-`k_T `Durham` algorithm, which forms the basis of our final results, we use the following configuration: `JetClustering::clustering_ee_kt(2, 4, 1, 0)`. This configuration is readily available within the official FCCAnalysis framework. Breaking down the arguments in `JetClustering::clustering_ee_kt`, the first argument indicates exclusive clustering, where the event is clustered into exactly n-jets. The second argument specifies the requirement of exactly four jets in the final state. The third argument denotes the energy ordering of the jets in the final state. Lastly, the fourth argument specifies the E recombination scheme.

The top quark decays through the dominant mode into a b-quark and a W-boson, with the W-boson subsequently decaying hadronically. Therefore, the final state of the FCNC process is characterized by the presence of 4 jets, out of which one or two may be b-tagged jets, and the missing transverse momentum should be less than 20 GeV to reduce the contribution of background events involving neutrinos.

We define three signal regions due to the varying number of required b-jets. In each of these regions, it is necessary for exactly one of the non-b-tagged jets to be classified as a light jet. All jets must have a transverse momentum of at least 20 GeV and a polar angle greater than 13 degrees. Additionally, events containing a lepton (either an electron or a muon) with a transverse momentum of at least 15 GeV are vetoed. In signal region 1, we select events with exactly 4 jets, none of which

are b-tagged. In signal region 2, we require exactly 4 jets, with one of them being b-tagged. In signal region 3, events with exactly 4 jets, with 2 of them being b-tagged, are selected. In cases where no b-tagged jets are present, we reconstruct top quark candidates based on the signal hypothesis. The kinematics of the top quark and W-boson candidates can be reconstructed from the corresponding decay particles by minimizing the following equation:

$$\chi^2 = (M_W - M^{\text{reco}}_{j_n j_m})^2 + (M_{\text{top}} - M^{\text{reco}}_{j_n j_m j_l})^2, \tag{2}$$

The equation represents the minimization of the chi-squared (χ^2) value for each signal region, where $M_{j_n j_m}$ denotes the reconstructed mass of the W system, and $M_{j_n j_m j_l}$ represents the reconstructed mass of the bW system. The process of minimizing the χ^2 value follows specific steps for each signal region. In signal region 1, we consider the jet permutation where any non-b-tagged jet can be assigned to j_n, j_m, j_l. This allows us to explore all possible combinations of assigning jets to the reconstructed particles and find the permutation that yields the minimum χ^2 value. In signal region 2, the jet permutation is considered such that any non-b-tagged jet can be assigned to either j_n or j_m, while j_l must correspond to a b-tagged jet. For signal region 3, any b-tagged jet can be assigned as j_l, and one of the remaining jets (j_n) must correspond to a non-b-tagged jet. The remaining jet (j_m) is then assigned accordingly. In Equation (1), the central values of $M_W = 80.37$ GeV and $M_{\text{top}} = 172.76$ GeV are considered for the masses of the W boson and the top quark, respectively. We use the same backgrounds listed in the previous section.

Based on the aforementioned selection, we performed the reconstruction of the W mass and top-quark mass distributions. These distributions are depicted in Figures 7 and 8. Specifically, the distributions showcase the $tc\gamma$ signal alongside the primary backgrounds at center-of-mass energies of $\sqrt{s} = 240$ GeV and $\sqrt{s} = 365$ GeV, respectively.

In addition to the W mass and top-quark mass distributions, we have utilized several other distributions that exhibit significant discrimination power for separating the signal from backgrounds. These distributions are presented in Fig 9, 10, and 11. These include the total momentum of the b-tagged jet, the light jet, the scalar sum of the transverse momentum of all jets in the final state, and the total momentum of the reconstructed top quark. In order to better discriminate signals from the background, we also examine the missing energy, and we define two new distributions as a ratio of the total momentum of the light jet to the b-tagged jet. We also use the ratio of the absolute value of momentum difference between the light jet and the b-tagged jets over the total momentum of b-tagged jets. During our fully-hadronic analysis, we have identified that the main contribution to the background arises from WW processes. To effectively suppress this background,

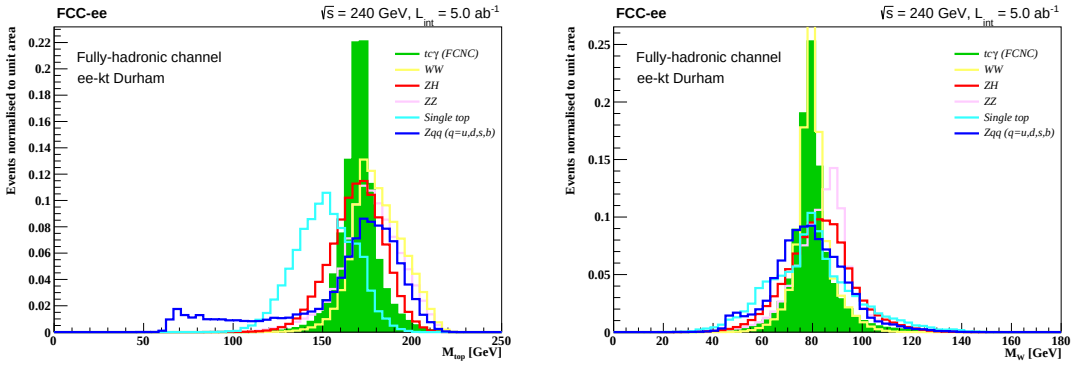

Figure 7: The mass distribution of the reconstructed top quark and the W boson for $tc\gamma$ signal in the presence of main backgrounds for the full-hadronic analysis. The distributions are shown for the $\sqrt{s} = 240$ GeV.

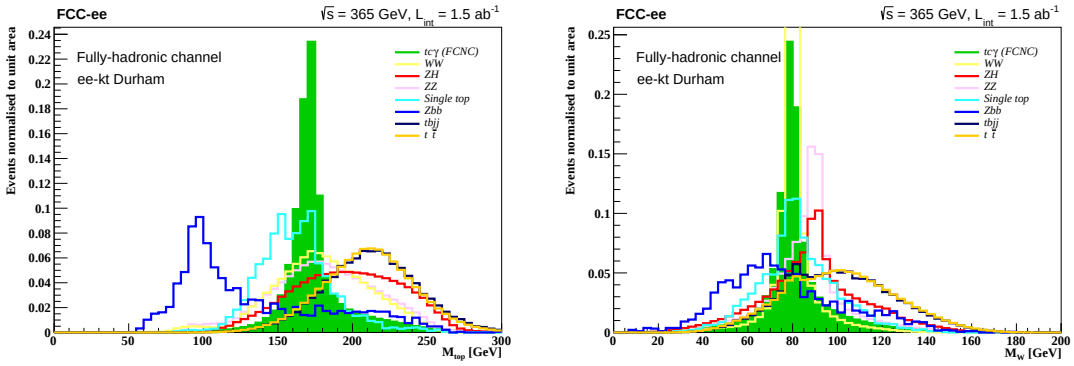

Figure 8: The mass distribution of the reconstructed top quark and the W boson for $tc\gamma$ signal in the presence of main backgrounds for the full-hadronic analysis. The distributions are shown for the $\sqrt{s} = 365$ GeV.

we utilize the reconstructed mass of the second W boson, which is present in the background but not in the signal. This mass is reconstructed using two jets that are not involved in the W boson reconstruction for the signal. In addition, we investigate the angular separations between the jets involved in the fully hadronic analysis, including both the b-tagged and light jets. Analyzing these angular separations helps enhance the discrimination between the signal and background. The

corresponding distributions for these angular separations can be found in Figure 11.

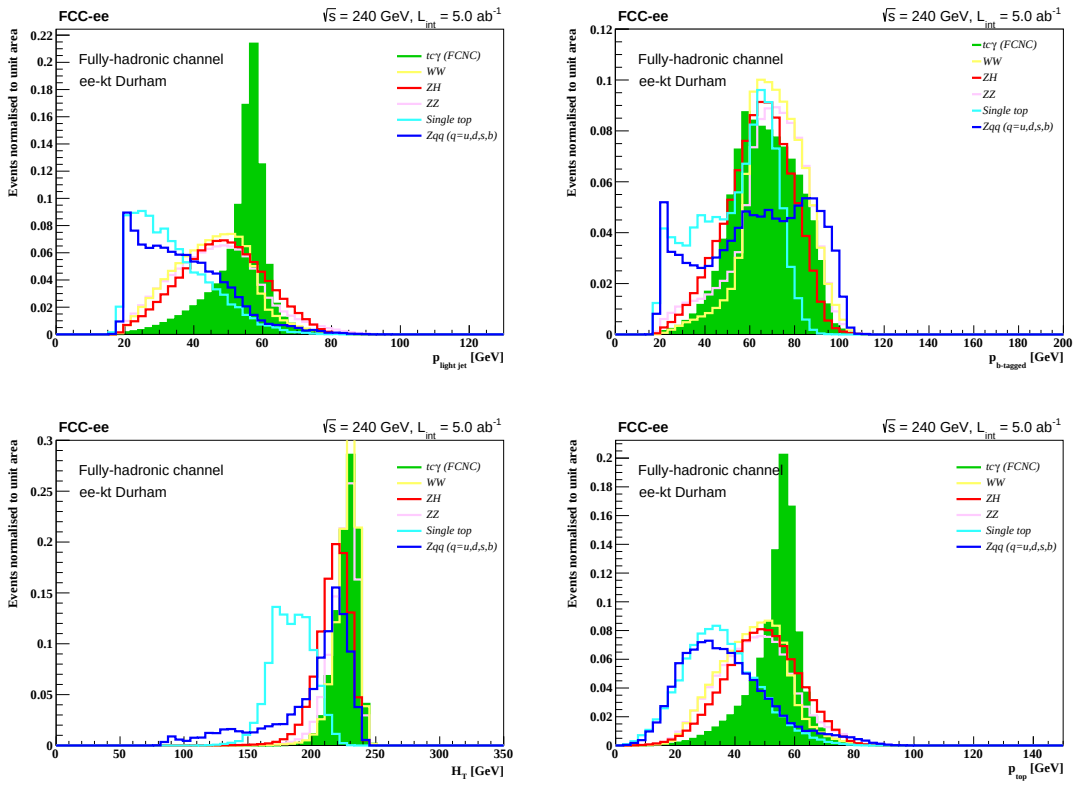

Figure 9: The total momentum of the b-tagged jets, the light jets, the scalar sum of the transverse momentum of all jets in the final state, and the total momentum of the reconstructed top quark for the $\sqrt{s} = 240$ GeV.

Based on the distribution presented above, we apply the following cuts to distinguish signals from backgrounds. The final selection cuts for the fully hadronic analysis at a center-of-mass energy of 240 GeV are as follows:

- The reconstructed top-quark mass distribution: 150 GeV $< M_{\text{top}} <$ 190 GeV.
- The reconstructed W boson mass distribution: 65 GeV $< M_{\text{W}} <$ 95 GeV.
- The momentum of the b-jet candidate: 40 GeV $< p_{\text{bjet}} <$ 95 GeV.
- The momentum of the highest energy light jet in each event: $p_{\text{lightjet}} >$ 50 GeV.

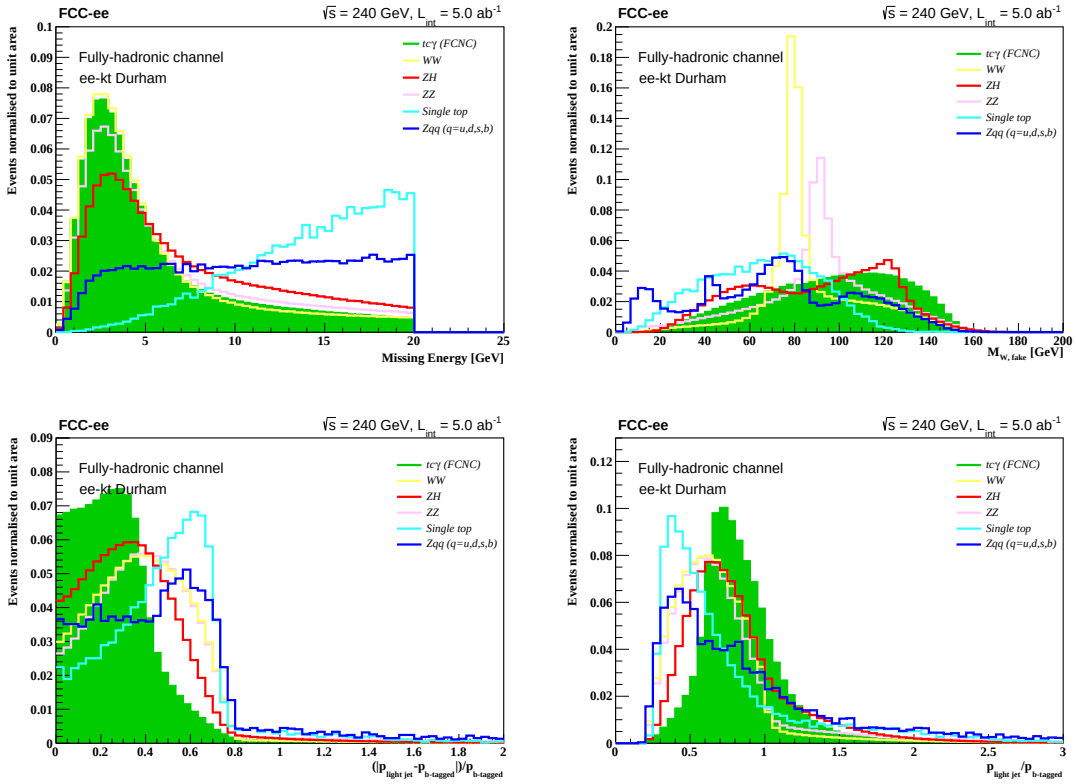

Figure 10: The distributions for the missing energy, the ratio of the total momentum of the light jet to the b-tagged jet, the ratio of the absolute value of momentum difference between the light jets and the b-tagged jets over the total momentum of the b-tagged jets, and the fake W boson mass distribution for the $\sqrt{s} = 240$ GeV.

- The momentum of the first and second jets used to reconstruct the fake W boson: 40 GeV $< p_{\mathrm{j1}} <$ 90 GeV and 25 GeV $< p_{\mathrm{j2}} <$ 65 GeV.
- Missing Energy (ME): $ME > 10$ GeV.
- The reconstructed invariant mass of the fake W boson: $M_{\mathrm{W}} < 70$ GeV or $M_{\mathrm{W}} > 100$ GeV.
- The maximum value of Missing Energy (ME): $ME < 10$ GeV.
- Angular separations between the jets: $\Delta R(j, j_2) > 1$, $0.5 < \Delta R(\text{light-jet}, \text{b-jet}) < 4.0$, $0.7 < \Delta R(\text{light-jet}, j_2) < 4.0$, $1.6 < \Delta R(\text{b-jet}, j_1) < 3.5$, $0.8 < \Delta R(\text{b-jet}, j_2) < 4.0$, and $0.7 < \Delta R(\text{light-jet}, j_1) < 3.5$.

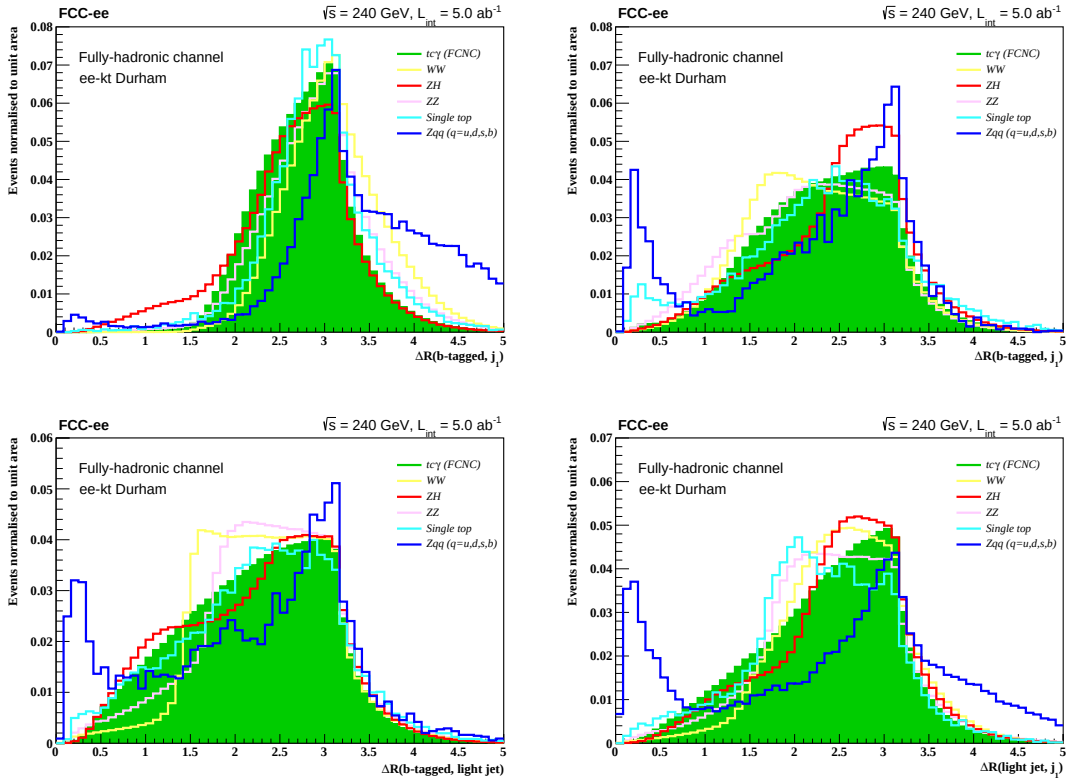

Figure 11: The angular separations between the jets (b-tagged and light jets) contributing to the fully hadronic analysis at the $\sqrt{s} = 240$ GeV.

- The ratio of the total momentum of the b-jet to the light jet: $0.60 < \frac{p_{\text{light-jet}}}{p_{\text{b-tagged}}} < 1.50$.
- The ratio of the momentum of the b-jet to the sum of the b-jet and light jet momenta: $\frac{p_{\text{b-jet}}}{p_{\text{b-jet}}+p_{\text{light-jet}}} < 0.50$.
- The scalar sum of momentum H_T: $H_T > 210$ GeV.

For the case of fully-hadronic analysis at 365 GeV, we consider the following selection cuts:

- Final selection cuts for the case of fully hadronic analysis at 240 GeV center-of-mass energy:
  - The reconstructed top-quark mass distribution 150 GeV $< M_{\text{top}} <$ 190 GeV.
  - The reconstructed W boson mass distribution 65 GeV $< M_{\text{W}} <$ 95 GeV.
  - The momentum of b jet candidate 35 GeV $< p_{\text{bjet}} <$ 140 GeV.

- The momentum of the highest energy light jet in each event 120 GeV $< p_{\text{lightjet}} <$ 150 GeV.
- The momentum of the first and second jets in each event that we use to reconstruct the fake W-boson, 40 GeV $< p_{\text{j1}} <$ 140 GeV and $p_{\text{j2}} >$ 80 GeV.
- The Missing Energy ME $>$ 10 GeV.
- The reconstructed invariant ass of fake W boson $M_{\text{W}} >$ 100 GeV.
- The maximum value of Missing Energy ME $<$ 10 GeV.
- The angular separation between the jets, i.e. $\Delta R(j, j_2) > 1$; $1.0 < \Delta R(light - jet, b - jet) < 3.5$; $0.7 < \Delta R(light - jet, j_2) < 3.5$; $1.0 < \Delta R(b - jet, j_1) < 3.2$; $0.5 < \Delta R(b - jet, j_2) < 3.5$; $1.7 < \Delta R(light - jet, j_1) < 3.5$.
- The scalar sum of momentum $H_T >$ 330 GeV.
- The total momentum of reconstructed top-quark, 110 GeV $< p^{\text{top}} <$ 150 GeV.

After applying the typical pre-selection and final-selection cuts mentioned above, we present the cross sections for the considered backgrounds at center-of-mass energies of 240 GeV and 365 GeV in Table VII and Table VIII, respectively.

$\sqrt{s}$ = 240 GeV	$tc\gamma$	tcZ	ZH	WW	ZZ	eebb	eeqq (q=uds)	single top
Cross section (fb)	28.39	67.44	201.868	16438.5	1358.99	455.7	1635.9	0.01075
Preselection	22.571	38.651	83.452	4285.272	400.682	0.2511	1.3863	0.00082
Final selection	6.609	11.320	8.361	257.030	34.193	0.00018	0.0027	4.83×10^{-7}

Table VII: Cross-sections in the unit of fb for the signal scenario with fully hadronic decay and for the main SM backgrounds considered in this study passing pre-selection and final selection cuts at a center-of-mass energy of 240 GeV.

$\sqrt{s}$ = 365 GeV	$tc\gamma$	tcZ	ZH	WW	ZZ	$t\bar{t}$	eebb	single top	$tbjj$
Cross section (fb)	52.39	104.0	117.3	10716.5	642.8	800.0	615.2	0.227	5.98
Preselection	39.152	76.435	32.779	2137.639	0.2038	271.443	0.1626	0.0034	2.973
Final selection	14.499	28.318	2.194	52.307	5.775	5.689	8.66×10^{-5}	~ 0	0.055

Table VIII: Same as Table. VII but this time for the center-of-mass energy of 365 GeV.

E. Results for the fully hadronic analysis

The primary results and findings from the fully hadronic analysis conducted at both 240 GeV and 365 GeV center-of-mass energies are highlighted in this section. Using the CLs method, the upper limits at the 95% confidence level (CL) for the top FCNC decays at a center-of-mass energy of $\sqrt{s} = 240$ GeV are determined. These limits are summarized in Table IX. These limits correspond to the FCC-ee operating with an integrated luminosity of 5 ab^{-1} at a center-of-mass energy of $\sqrt{s} = 240$ GeV. The corresponding values for $\sqrt{s} = 365$ GeV, along with an integrated luminosity of 1.5 ab^{-1}, are provided in Table X.

$\sqrt{s} = 240$ GeV	$Br(t \to c\gamma)$	$Br(t \to cZ)$
Full hadronic decay	1.98×10^{-4}	1.01×10^{-4}

Table IX: The upper limits on the top FCNC decays for the fully hadronic decay analysis at 95% C.L obtained using the CLs method at the $\sqrt{s} = 240$ GeV and for an integrated luminosity of 5 ab^{-1}.

$\sqrt{s} = 365$ GeV	$Br(t \to c\gamma)$	$Br(t \to cZ)$
Full hadronic decay	8.45×10^{-5}	3.70×10^{-5}

Table X: Same as Table. IV, but this time for the $\sqrt{s} = 365$ GeV and for an integrated luminosity of 1.5 ab^{-1}.

F. Statistical combination of results in hadronic channel

In Table. XI, the upper limits on the top FCNC decays at 95% C.L extracted using the `Higgs Combine Tool` [20] after the statistical combination of $\sqrt{s} = 240$ and 365 GeV are presented.

$\sqrt{s} = 240$ and 365 GeV	$Br(t \to c\gamma)$	$Br(t \to cZ)$
Statistical combination	7.65×10^{-5}	3.28×10^{-5}

Table XI: The upper limits on the top FCNC decays at 95% C.L obtained using the Higgs combination tools after combination of $\sqrt{s} = 240$ and 365 GeV.

V. DISCUSSION AND CONCLUSION

It is anticipated that the future high-energy and high-luminosity collider, FCC-ee, with its clean environment and high luminosity, will offer a unique opportunity to accurately measure the properties of top quarks and their interactions. In this study, we investigated the potential sensitivity of FCC-ee to flavor-changing neutral current (FCNC) transitions involving top quarks. Specifically, we focused on the tqZ and $tq\gamma$ couplings, where $q = u, c$. To ensure model independence, we employed the effective Lagrangian approach and examined the single top-quark production process in e^-e^+ collisions, specifically $e^-e^+ \to \bar{t}q + t\bar{q}$. Our analysis considered both leptonic decays (electron and muon) and fully hadronic decays of the W boson in the top quark decay. To conduct our study, we utilized the official FCCAnalysis framework and employed various jet clustering algorithms. We generated FCNC signal and background samples using the IDEA detector response, and also generated samples in the EDM4HEP format. Through our analysis, we demonstrated that by combining energy and channels, and considering integrated luminosities of 5 and 1.5 ab^{-1}, we could achieve sensitivity on the order of 10^{-5} to 10^{-6}.

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[1] F. Abe *et al.* [CDF], "Observation of top quark production in $\bar{p}p$ collisions," Phys. Rev. Lett. **74**, 2626-2631 (1995) doi:10.1103/PhysRevLett.74.2626 [arXiv:hep-ex/9503002 [hep-ex]].

[2] S. Abachi *et al.* [D0], "Observation of the top quark," Phys. Rev. Lett. **74**, 2632-2637 (1995) doi:10.1103/PhysRevLett.74.2632 [arXiv:hep-ex/9503003 [hep-ex]].

[3] S. L. Glashow, J. Iliopoulos and L. Maiani, "Weak Interactions with Lepton-Hadron Symmetry," Phys. Rev. D **2**, 1285-1292 (1970) doi:10.1103/PhysRevD.2.1285

[4] J. A. Aguilar-Saavedra, "Top flavor-changing neutral interactions: Theoretical expectations and experimental detection," Acta Phys. Polon. B **35**, 2695-2710 (2004) [arXiv:hep-ph/0409342 [hep-ph]].

[5] J. A. Aguilar-Saavedra, "A Minimal set of top anomalous couplings," Nucl. Phys. B **812**, 181-204 (2009) doi:10.1016/j.nuclphysb.2008.12.012 [arXiv:0811.3842 [hep-ph]].

[6] A. Alloul, N. D. Christensen, C. Degrande, C. Duhr and B. Fuks, "FeynRules 2.0 - A complete toolbox for tree-level phenomenology," Comput. Phys. Commun. **185**, 2250-2300 (2014), [arXiv:1310.1921 [hep-ph]].

[7] C. Duhr and B. Fuks, "A superspace module for the FeynRules package," Comput. Phys. Commun. **182**, 2404-2426 (2011), [arXiv:1102.4191 [hep-ph]].

[8] N. D. Christensen, P. de Aquino, C. Degrande, C. Duhr, B. Fuks, M. Herquet, F. Maltoni and S. Schumann, "A Comprehensive approach to new physics simulations," Eur. Phys. J. C **71**, 1541 (2011), [arXiv:0906.2474 [hep-ph]].

[9] C. Degrande, C. Duhr, B. Fuks, D. Grellscheid, O. Mattelaer and T. Reiter, "UFO - The Universal FeynRules Output," Comput. Phys. Commun. **183**, 1201-1214 (2012), [arXiv:1108.2040 [hep-ph]].

[10] S. Mrenna and P. Skands, Phys. Rev. D **94**, no.7, 074005 (2016) doi:10.1103/PhysRevD.94.074005 [arXiv:1605.08352 [hep-ph]].

[11] J. de Favereau *et al.* [DELPHES 3], JHEP **02**, 057 (2014) doi:10.1007/JHEP02(2014)057 [arXiv:1307.6346 [hep-ex]].

[12] FCC-ee IDEA detector model, Elisa Fontanesi, Lorenzo Pezzotti, Massimiliano Antonello, Michele Selvaggi, https://indico.cern.ch/event/1173562/contributions/4929025/attachments/2470068/4237859/2022-06-FCC-jets.pdf

[13] S. Catani, Y. L. Dokshitzer, M. Olsson, G. Turnock and B. R. Webber, "New clustering algorithm for multi - jet cross-sections in e+ e- annihilation," Phys. Lett. B **269**, 432-438 (1991),

[14] P. Z. Lai, M. Ruan and C. M. Kuo, "Jet performance at the circular electron-positron collider," JINST **16**, no.07, P07037 (2021), [arXiv:2104.05029 [hep-ex]].

[15] F. Bedeschi, L. Gouskos and M. Selvaggi, "Jet flavour tagging for future colliders with fast simulation," Eur. Phys. J. C **82**, no.7, 646 (2022), [arXiv:2202.03285 [hep-ex]].

[16] J. A. Aguilar-Saavedra, "Top flavor-changing neutral interactions: Theoretical expectations and experimental detection," Acta Phys. Polon. B **35**, 2695-2710 (2004),

[17] J. A. Aguilar-Saavedra, "A Minimal set of top anomalous couplings," Nucl. Phys. B **812**, 181-204 (2009), [arXiv:0811.3842 [hep-ph]].

[18] J. A. Aguilar-Saavedra, "A Minimal set of top-Higgs anomalous couplings," Nucl. Phys. B **821**, 215-227 (2009), [arXiv:0904.2387 [hep-ph]].

[19] R. Tenchini and C. Verzegnassi, *The Physics of the W and Z Bosons*, World Scientific, 2007.

[20] https://cms-analysis.github.io/HiggsAnalysis-CombinedLimit/